

Chapter 11

Grammaticalization of future markers in Thai

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Contemporary Thai has a number of forms that mark the notion of futurity. The primary future tense marker is *cà*, whose lexical origin, though not yet established, is likely the verb *càk* 'know'. Other markers are used either alone or in combination with *cà(k)*. Their morphosyntactic development reveals some influence of the typological and idiosyncratic features of Thai, such as pragmatic orientation, verb serialization, and preference for polylexemic units in lexicalization and grammaticalization, among others. Their semantic developments are also well motivated in terms of conceptual relatedness between the source forms and the grammaticalized forms. A review of grammaticalization scenarios with reference to grammaticalization parameters, e.g., desemanticization, extension, decategorialization, and erosion, shows that the Thai future markers have been grammaticalized to varying degrees.

1 Introduction

Living in the spatio-temporal dimension, all humans supposedly have linguistic means, be they lexical or grammatical, of expressing future time reference.¹ As observed in the literature, Thai does not have an inflectional future, as a general characteristic of an isolating language, but rather has a construction involving the auxiliary *cà* combining with a verb to indicate future time reference (Dahl 1985: 173–174).² Thai also has a handful of other markers, with variable degrees

¹For instance, Dahl (1985: 108–109) notes that even in the languages without a future tense marker, there are ways of signaling future time reference.

²Thai linguists use a few different romanization systems. We use the Thai IPA transcription.



of grammaticalization, that signal that an event or state of affairs will occur or exist at a future time. Their developmental patterns exhibit certain commonalities with other languages as well as peculiarities due to the typological or idiosyncratic features of Thai. This chapter describes the means of future time reference in Thai from a grammaticalization perspective. The data sources include the Thai National Corpus (TNC), dictionaries, lexica, online resources, and reference grammars.³

Thai is a Kra-Dai language (Ostapirat 2000; Eberhard et al. 2022), also known as Tai-Kadai (Diller et al. 2008), and is mostly spoken by the people in Thailand and its vicinity. Thai is an isolating language. Word order is SVO and phrases are head-initial. Thai does not have case marking morphology, but it has a complex tone system (for typological descriptions, see Noss 1954, 1964; Abramson 1962; Haas 1964; Anthony et al. 1968; Warotamasikkhadit 1972; Tingsabadh & Abramson 1999; Campbell 2000; among others).

This chapter is organized as follows: Section 2 describes and exemplifies a few forms indicating future time reference, i.e., *càk* (Section 2.1), *cà* (Section 2.2), adverbials (Section 2.3), and particles (Section 2.4); Section 3 addresses some prominent issues with respect to grammaticalization of future time reference, focusing on the morphosyntactic motivation (Section 3.1), the conceptual motivation (Section 3.2), and grammaticalization parameters (Section 3.3); and Section 4 summarizes the findings and concludes the chapter.

2 Future time reference in Thai

Some Thai reference grammars explicitly state that Thai has no future tense markers, largely because their use is not mandatory and future time reference can be inferred from a temporal adverb, an aspect marker, or from the context (Iwasaki & Ingkaphirom 2005: 123; Thiengburanathum 2013: 159). However, most reference grammars acknowledge the presence of future markers (Noss 1964: 165; Campbell & Shaweevongse 1972: 26; Setthapun 1992: 24; Smyth 2002: 67; among others). The most commonly cited future marker is *cà*, but there are other forms as well. The notion of futurity signaled by certain words is often interpreted as a lexical meaning rather than a grammatical function. This is the case because of categorial fluidity, resistance to formal change, as well as the prevalence of the polyfunctionality of forms across lexical and grammatical domains, which are characteristics typical of isolating languages.

³The Thai National Corpus (TNC) is an online searchable, 33.4-million-word corpus, developed by Chulalongkorn University.

However, if we adopt a dynamic perspective such as the one in grammaticalization theory, Thai indeed has a number markers of futurity. They include *càk*, *cà*, *cuan*, *kùap*, *thêep*, and *klây*. There are poly-lexemic forms created by means of compounding. Some authors, e.g., Supanvanich (1973), also claim that certain sentence ending particles in Thai function as markers of future time reference. In the following, we illustrate the uses of such future markers.

2.1 *Càk*

The word *càk* marks the future in Modern Thai. Evidently, it is the earlier form of *cà* (see Section 3.2 below), the primary future marker in Contemporary Thai. Following the tenets of grammaticalization theory according to which all grammatical forms originate from a lexical source, *càk* is also believed to have a lexical origin. With a regrettably shallow history of documentation (see Section 3.3), however, there has not been any serious scholarly investigation to date into its origin, and, consequently, there is no consensus regarding its lexical origin. Despite the paucity of lexical evidence, some clues do exist in language contact. For instance, the Thai lexicon was greatly influenced by Middle Chinese, and a lexicon of Middle Chinese borrowings (SEALANG 2006) includes Thai *จັก* *càk* (*jàk* in its notation). As the historical source of the Thai *càk*, this lexicon points to the Middle Chinese lexeme denoting ‘recognize (as is)’, which references Proto-Southwestern Tai **cak* ‘to recognize, know’ (Jonsson 1991), as well as Proto-Southwestern Tai **čak* and Proto-Tai **čak* (Li 1977). The same source lists the Thai verb *รู้จັก* *rúucàk* (= *rujàk*) ‘to recognize, know’ as a borrowing of the Middle Chinese lexeme 識 (*śik*) ‘to recognize, know, discern, have knowledge, etc.’, citing Manomaivibool (1975; known as ‘Prapin’), Baxter (2000), and Starostin (n.d.) (SEALANG 2006). Based on these comparative-linguistic analyses, it can be hypothesized that the future marker *càk* (and its later form *cà*, see Section 2.2 below) developed from the Middle Chinese borrowing *càk* with the meaning ‘know’.⁴ Despite *càk* being used as the cognitive verb ‘to know, under-

⁴Other possibilities from the data in SEALANG are the two homophonous lexemes (*จັก* *càk*; *jàk* in its notation), one denoting ‘weave, knit’ (Middle Chinese *čík/tsyik*) and the other denoting ‘split, pare bamboo’ (Proto Southwestern **cak/*čak* and Proto-Tai **čak*), the latter being proposed in the earlier version of this paper. An anonymous reviewer raised concerns that the proposal is not substantiated by historical data. Dr. Weera Ostapirat (p.c.), a specialist in the Tai language family, also expresses his opinion that ‘split, pare bamboo’ is not a likely source and that ‘know’ is the better candidate (see Section 3.2 for conceptual motivation across languages). He further states that the native Thai verb *càak* ‘depart, leave’ is a less likely but possible candidate. At the current level of understanding, the lexical source of the future marker *cà(k)* remains inconclusive. Special thanks go to the reviewer and Dr. Weera Ostapirat for their suggestions.

stand, be aware of', it has become obsolete and only survives in the form of *rúucàk* 'to know a person, place, etc.' (thus, recognition-oriented), in which *rúu* also means 'to know a subject or piece of learned information' (thai-language.com) (thus, comprehension-oriented).⁵ The use of the reinforced *rúucàk* is exemplified in (1):⁶

(1) Modern Thai (constructed)

ผู้ชาย ที่ ผึ้ง รู้จัก ใน เมือง เชียงใหม่ ก็ มี อยู่ สอง คน
phûuchaay thîi phûng rúucàk nay muan Chiang.mây kôw mii yûu sǎw
man that [name] know in city [name] also have be two
khon
person

'There are two men whom Pheung knows in Chiang Mai.'

(Modern Thai, constructed)

Càk has been grammaticalized as a future auxiliary (for morphosyntactic and conceptual motivation see Section 3.1 and 3.2). As an auxiliary, first attested in the Sukhothai period (1238–1347 CE), *càk* indicates future or intention.⁷ Its use is exemplified below:

⁵It has been observed that Thai has an idiosyncratic tendency of favoring a morphological process of 'recycling', whereby synonyms or near synonyms are compounded. The multiple forms thus created tend to create 'layering' (Hopper 1991) and have different degrees of intensification or different register-specific, semantic specialization (see below for more instances). It is also common that there are no noticeable semantic differences between the multiple forms created through recycling. Some of such (near-)synonymous forms may disappear, as shown here with *càk* 'to know'.

⁶Note that Thai does not practice inter-lexical spacing, but Thai data are given with spacing for visual clarity.

⁷In the excerpt, the parts that the original Thai data do not need to be presented with interlinear morphemic glossing are presented in English translation within square brackets in the interest of space. In the English translation, the parts that are required for syntactic and semantic reasons but are only implicit in the original data are presented within parentheses. An alternative translation is presented in parentheses as well. Also for convenience, markers of future time, regardless of their grammatical status or additional semantics, are glossed as FUT(ure), unless a different label is needed for expositional clarity.

(2) *Càk* future/intention

- a. ใคร จัก ใคร่ คำ ช้าง คำ ใคร จัก ใคร่ คำ ม้า คำ ใคร จัก ใคร่ คำ เงิน คำ ทอง คำ

khray càk khây kháa cháaŋ kháa khray càk khây kháa máa
 who FUT want sell elephant sell who FUT want sell horse
kháa khray càk khây kháa ɸuan kháa thooŋ kháa
 sell who FUT want sell silver sell gold sell

‘Whoever would want to trade in elephants does so. Whoever would want to trade in horses does so. Whoever would want to trade in silver or gold does so.’

(1292, King Ramkhamhaeng’s free trade law)

- b. *Càk* future/intention

เมื่อ ไม่ รัก จัก หย่า ก็ ว่า เลิกล

mêu mây rák càk yàa kôw wâa thòet
 when not love FUT divorce so say please

‘When (you) do not love and (you) would (intend to) divorce (me), please tell (me).’

(2009, Enak Phra Khanong)

Excerpt (2a) is taken from a historical text, in which *càk* functions as a future marker with the main verb *khây* ‘want’. While *càk* can mean either future or intention (the notion of intention may have arisen from the knowledge meanings associated with the source verb), the presence of the co-occurring conative verb *khây* ‘want’ makes the intention interpretation redundant and the future interpretation contextually favored. Excerpt (2b) is taken from a Modern Thai source, part of a traditional Thai octameter poem entitled ‘The Legend of Mae Nak Phra Khanong’, which still bears many characteristic features of the classical poem. In this excerpt, *càk* can mean either future or intention. One might argue that if the phrase *càk yàa* ‘will divorce/intend to divorce’ is viewed in syntactic parallelism with the preceding *mây rák* ‘not love’, *càk* would favor the intention interpretation, i.e., as a main verb, rather than the future interpretation, i.e., as an auxiliary. However, since Thai does not require syntactic parallelism, there is no syntactically-driven interpretational favoring. The future interpretation is as much favored as in (2a), especially because the second predicate (i.e., ‘divorce’) is later in time (thus, future) relative to the first predicate (i.e., ‘not love’).

The future auxiliary *càk* can form a deontic modal auxiliary with *tôŋ* in the form of *càk-tôŋ* ‘must’, especially in formal registers, e.g., legal documents. As

briefly alluded to before, ‘recycling’ a formant for use in combination with another formant is a common lexicalization-grammaticalization strategy in Thai (Rhee & Khammee 2022, Khammee & Rhee 2022, Thirakunkovit & Rhee 2023; see also Section 3.3.4). The participating form *tôŋ* is itself the deontic modal ‘must’, which is also heterosemous (Lichtenberk 1991) with the verb *tôŋ* ‘to make contact, be exposed to, touch, agree with, etc.’

- (3) *càk-tôŋ* future/deontic necessity
 ความ ยุติธรรม จักต้อง ดำรง อยู่ ใน มวล มนุษย์
khwaam yútitham càk-tôŋ damron yuu nay muan manút
 NMLZ justice FUT maintain be in entire human
 ‘Justice will (shall, must) prevail in mankind.’
 (Modern Thai, constructed)

Excerpt (3) contains *càk-tôŋ*, a future marker also with a deontic necessity meaning. An interesting aspect is that, due to the fact that deonticity is inherently future-related, the presence of the future marker *càk* is not semantically consequential, i.e., there is no perceivable difference between *càk-tôŋ* and *tôŋ*, both denoting ‘must’. The close relationship between deontic modality (agent-oriented) and future tense has been observed in Fleischman (1982), Dahl (1985), Heine (1993), Bybee et al. (1994), among others.

2.2 Cà

In Modern Thai, the primary future marker is *cà*, which is historically an eroded form of *càk*, addressed in the previous section.⁸ The future-marking function of *cà* is first identified in the Ayutthaya Period (1351–1767). In the earliest record of *cà*, the form strongly suggests intention, thus often making the sentence ambiguous as to futurity or intention (note that the two are conceptually close, as exemplified in the development of the English future *will*, Aijmer 1985: 11–20).⁹ As indicated previously, it is quite possible that *cà*, from its earlier verbal meanings of *càk* ‘to know’, has developed into ‘to intend’, and its conceptually close ‘future’ meaning has emerged simultaneously or soon afterwards. Thus, even from the early usage, *cà* functions as a marker not only of the future but also of

⁸The older *càk* is now used with very low frequency (52 pmw in a contemporary corpus TNC) and functions as an auxiliary, marking the future tense or the subject’s intention. On the other hand, the newer and eroded *cà* occurs with very high frequency as a general future-tense marker (15,014 pmw in TNC).

⁹Soithurum (2010), conceding that the absolute lexical origin of *cà* is unavailable, asserts that its earliest recognizable meaning is ‘intend’.

intention and even desire, and other meanings such as assurance and obligation become associated with *cà*, as these notions are all closely related to futurity (cf. Dahl 1985: 103; also see below). The following is an excerpt involving the future marker *cà*:

(4) *cà* future/intention

(in a letter by the Siamese ambassador to France)

จะ ขอบ แก่ ท่าน ให้ เข้า ทูล แก่ พระมหากษัตริย์ เจ้า ผู้ ไหวญ ...

cà khǎw kèe thân hây khâw thuun kèe phrámahākàsàt-câaw phûu
FUT beg to you:HON to enter tell to king who

yà-y [...]*lêe cà ù-sàa thamhây cháná thúkkhon nay khwaam*
great and FUT attempt make win everybody in NMLZ

nóp-nǎw nán

reverence that

‘We would (intend to) beg Your Excellency to tell His Majesty [...] and (we) will (intend to) endeavor to surpass everybody else in terms of respect and veneration toward the king.’

(Siamese Documents of the 17th Century, Paris)

Excerpt (4) is taken from a letter dated from the 17th century, in which the Siamese ambassador requests the addressee to convey a message of allegiance and veneration to the French king. As seen in the excerpt, the future marker *cà*, which, in addition to being a future auxiliary, has potential as a verb, and is followed by three successive verbs *ù-sàa* ‘attempt’, *thamhây* ‘make’, and *cháná* ‘win’.¹⁰

The auxiliary *cà* denotes, in addition to future tense, various future-related modal notions, e.g., intention, promise, necessity, possibility, probability, certainty, indeterminacy, proximation, constancy, etc., either alone or through ‘univerbation’ (Lehmann 2015) with other lexemes.¹¹ These notions are so closely related that whichever is denoted by this particular instance of a *cà*-sentence largely depends on the context, and functional designation may not be exclusive. The usages of *cà* for denoting these modal notions are exemplified below:

¹⁰This is a unique situation in Thai, where multiple verbs may occur in succession asyndetically. For instance, Diller (2006: 160) notes that Thai, which he called ‘a verb-loving language’, allows such series of verbs as ‘Peel give eat’ (for ‘Peel the oranges for the guests’), ‘Tell finish not have’ (for ‘I already told you, there aren’t any’), and the like. See 3.1 for more discussion on its relevance to grammaticalization.

¹¹Since modal markers in Thai are many and varied, with tense and aspect markers omissible without affecting grammaticality, Soithurum (2010: 114) asserts that Thai is not a tense- or aspect-prominent language but a ‘mood-prominent’ language.

- (5) a. *cà* future/intention
 ฉัน จะ เข้าร่วม ประชุม ที่ เยอรมนี เดือน หน้า
chǎn cà khâwruam pràchum thii yəəramánii dʰan nâa
 I FUT attend meeting at Germany month next
 'I will (intend to) join a meeting in Germany next month.'
 (Modern Thai, constructed)
- b. *cà* future/promise/determination
 ฉัน จะ ไม่ ซื้อ ของ ร้าน นี้ อีก ต่ไป
chǎn cà mây sùu khwǎn ráan níi iik tɔ̀wpay
 I FUT not buy thing shop this again further
 'I will (promise, am determined) not (to) buy any stuff again from this shop.'
 (Modern Thai, constructed)
- c. *cà* future/prediction
 อากาศ ที่ ปลอดโปร่ง จะ ทำให้ เรา พ้นตัว จาก ไข้ เร็ว ยิ่ง ขึ้น
aakàat thii plwɔ̀tpròɔŋ cà thamhây raw fɨ́ntua càak khây rew
 air that clean FUT make us recover from fever fast
yîŋ khûn
 more up
 'Clean air will (is expected to) help us recover from a fever faster.'
 (Modern Thai, constructed)
- d. *cà* future/irrealis
 ถ้าผม จะ เปลี่ยนอาชีพ ผม จะ เป็น ครู
thâa phǒm cà plian aachîip phǒm cà pen khruu
 If 1SG:M FUT change occupation 1SG:M FUT be teacher
 'If I had changed careers, I would be a teacher (now).'
 (Modern Thai, constructed)

In (5), *cà* signals that the events described belong to a point in time in the future and at the same time implies certain concepts typically associated with the future tense, as indicated above. Evidently, in (5a), the agent of the predicate with an action verb in the active voice (i.e., attend a meeting) is the sentential subject, which coincides with the speaker (i.e., 'I'), an ideal condition for invoking the speaker's intention. In (5b), the semantic-syntactic features are similar to those of (5a), with an added feature of negation of intention of any further patronage of the shop, which lends the interpretation of determination and promise. In (5c), the sentential subject is non-sentient (i.e., air), thus not amenable to any interpretation involving conative modality such as intention, promise, determination,

etc. The pragmatic inferences, based on the scientific knowledge of the causal relationship between health benefit and clean air, tend to suggest a prediction. Finally, as in (5d), if the auxiliary *cà* is used in the apodosis or protasis of a conditional sentence, especially of a hypothetical condition, it functions as an irrealis marker (cf. ‘hypothetical situation or putative action’ Noss 1964: 165; Soithurum 2010). This is in line with the observation that future is commonly associated with a prediction or extrapolation from the present state of the world (Dahl 1985: 103; Comrie 1989; Fleischman 1982: 24–31, 133–134; see also Kratzer 1991; Sarkar 1998; Sansó 2006: 243; Soithurum 2010; Aikhenwald 2010: 81; Stenzel & Gomez-Imbert 2018: 362, for individual instances).

In addition to the usages of *cà* exemplified above, *cà* also signals diverse future-related modal notions, e.g., necessity, indeterminacy, possibility, certainty, proximation, constancy, etc., through univerbation with other lexemes, which contribute additional meanings of futurity. In this context, the univerbation phenomenon, frequently observed in Thai, warrants a brief exposition. When *cà* forms a univerbated unit with another form (typically a modal), it is possible to treat the two modals in a complex form as operating separately, instead of treating the form as a single unit.¹² For instance, *cà* and *tôŋ* (see 6a below) each may be analyzed as signaling future and deontic necessity. This is a perfectly valid approach, especially considering that when *cà* is omitted from *cà-tôŋ*, there may not be loss of deontic necessity (obligation) meaning, and conversely when *tôŋ* is omitted from *cà-tôŋ*, there may not be loss of futurity meaning. Such an analytic approach, however, does not capture the insights of the researchers who acknowledge the patterns of common co-occurrence of *cà* and other modal elements, and therefore present them as more or less combined forms (even though *cà* may be optional in certain cases). For instance, Noss (1964: 117), Smyth (2002: 65–69), Iwasaki & Ingkaphirom (2005: 130–134), and Soithurum (2010: 58), among others, present certain combinations as unitized forms, either explicitly or implicitly, e.g., (*cà*) *tôŋ* ‘must’, *khuan* (*cà*) ‘should/ought’, *nâa cà* ‘should/ought’, *yàak* (*cà*) ‘want to, would like to’, *campen cà* ‘necessary to’, *kamlan cà* ‘be about to (happen)’, *kùap* (*cà*) ‘be about to (happen)’, *thêep* (*cà*) ‘be about to (happen)’, *cuan* (*cà*) ‘be about to (happen)’, *mák cà* ‘used to, would’, *yôm cà* ‘likely’, *chák cà* ‘beginning to’, *nâa cà* ‘possibly’, *khon cà* ‘probably’, *àat cà* ‘probably, perhaps’, etc. Therefore, while the treatment of such forms either analytically or holistically may be an unnecessarily granular issue, the fact that they tend to accompany other forms and that such a tendency is not applicable to all modal elements supports the practice of treating them as more or less unitized forms (note that the combined forms are represented with a hyphen between the two components).

¹²Our thanks go to the anonymous reviewer for raising this issue.

The usages of the *cà*-univerbated forms for signaling diverse future-related modal notions are exemplified in the following. The first set involves necessity as in (6):

- (6) a. *cà-tôṇ* future/necessity
 คุณ จะต้อง จ่าย ค่าปรับ ที่ สถานี ตำรวจ
khun cà-tôṇ càay khâapràp thii sathǎanii tamrùat
 you FUT pay fine at station police
 ‘You will (must, have to) pay the fine at the police station.’
 (Modern Thai, constructed)
- b. *khuan-cà* future/necessity
 เธอ เปียก ทุกคน เลย เธอ ควรจะ พก ร่ม มา ด้วย
thəə piak mət ləəy thəə khuan-cà phók rôm maa dûay
 you wet all already you FUT bring umbrella come together
 ‘You are all wet. You should have brought an umbrella.’
 (Modern Thai, constructed)

Example (6a) above illustrates the usage of the future auxiliary *cà-tôṇ*, which also signals necessity. The additional morpheme *tôṇ* is itself a deontic auxiliary, a heteroseme of the verb *tôṇ* ‘to touch, be exposed to, agree’ (see Excerpt 3 of *càk-tôṇ* in Section 2.1 above). Since the phonologically longer form *càk-tôṇ* and the shorter form *cà-tôṇ* carry the identical function, it can be assumed that *càk-tôṇ* and *cà-tôṇ* are variants and that the latter is the more grammaticalized of the two (see Section 3.3.4 below). Similarly, (6b) exemplifies the use of *khuan-cà*, an auxiliary for future and necessity, which denotes unfulfilled obligation. This compound incorporates the future *cà* and the necessity modal auxiliary *khuan*. Even though *cà-tôṇ* and *khuan-cà* both denote future and necessity, the former is stronger in terms of the degree of necessity than the latter, analogous to the English deontic modals *must* and *should*. Another subtlety is that *cà-tôṇ* is more strongly future-oriented than *khuan-cà*, and thus in regret contexts as in (6b), *khuan-cà* is more appropriate.

As shown above, the position of *cà* in the combined forms varies (e.g., *cà-tôṇ* and *khuan-cà*).¹³ It is generally accepted that the relative position of a form with reference to another typically carries the meaning-changing potential due to the scope effect, as in the famous contrast between *ino-rssu-anguag* [person-big-little] ‘little giant’ and *inu-ngua-rssuag* [person-little-big] ‘big midget’ in a dialect

¹³ An anonymous reviewer raised this interesting issue, which may be related to scope differences.

of Eskimo (Sadock & Olsen 1976, as cited in Bybee 1985: 96). Soithurum (2010) identifies three different structural patterns of *cà*: (i) before a verb, (ii) before or after an auxiliary, and (iii) after a complementizer (*wâa* or *thîi*), and briefly mentions, using the example of *kamlan cà*, the structural scope of *kamlan* and *cà* (Soithurum 2010: 72, 76). The combinations described in the present analysis all belong to (ii). While Soithurum's classification system is useful, it is too imprecise to differentiate the types of, e.g., *cà-tôṅ* and *khuan-cà*, and does not adequately explain scope differences. Considering that *tôṅ* and *khuan*, for example, are both full-fledged auxiliaries of necessity, and that the relative position of *cà* in *cà-tôṅ* and *khuan-cà* cannot be reversed (**tôṅ-cà* and **cà-khuan*), the fixed relative position of *cà* is a great puzzle. It is possible that this frozen order represents the syntagmatic arrangement of the forms at the time of grammaticalization (cf. Givón 1971: 413). With the current paucity of relevant data, however, the issue requires further research.

The second set involves modals denoting variable degrees of certainty.

- (7) a. *nâa-cà* future/probability ('probably')

ทุกอย่าง น่าจะ จบลง ดีด้วย ดี
thúkyàaṅ nâa-cà còp.ləṅ dūay dii
 everything FUT end by good
 'Everything should end well.'

(Soithurum 2010: 79, modified)

- b. *àat-cà* future/possibility ('possibly')

ครู อาจจะ สอน อยู่ ที่ ห้องเรียน สีแดง ตอนบ่าย พรุ่งนี้
khruu àat-cà sǎwṇ yùu thîi hōṅrian sǐdēeṅ tɔwṇbàay
 teacher FUT teach PROG at classroom red.color afternoon
phrûṅnîi
 tomorrow

'The teacher may teach in the red classroom tomorrow afternoon.'

(Modern Thai, constructed)

- c. *khon-cà* future/possibility ('possibly')

คุณตา คงจะ ปลูก พืช อยู่ หลังบ้าน เย็น นี้
khuntaa khon-cà plùuk phàk yùu lǎṅbâan yen nîi
 grandpa FUT plant vegetable at backyard evening this

'Grandpa may have planted vegetables at the backyard this evening.'

(Modern Thai, constructed)

Example (7a) illustrates the use of *nâa-cà* denoting future and probability. The added form *nâa* is a modal adverb/auxiliary denoting possibility, which is inher-

ent in future, and vice versa. The conceptual affinity may have motivated this univerbation process. As a future marker, *nâa-cà* also indicates indeterminacy or uncertainty; thus, the described event is lower on the scale of certainty than *cà-tôη* derived from E-Necessity (high probability). The marker is also used in a regret situation, i.e., in describing an event that was once possible but was missed. Example (7b), involving *àat-cà*, in which *àat* is also a modal adverb/auxiliary, indicates likelihood.¹⁴ Its degree of certainty is also lower than probability, as signaled by *cà-tôη*. Example (7c), involving *khon-cà*, in which *khon* is a modal adverb/auxiliary of possibility, still bears a semantic connection with its adjectival heteroseme *khon* ‘be permanent, fixed’. The epistemic meaning (E-Possibility) of *khon-cà* assigns the word to be used as an evidential marker, as in (7c). In this context, variability of the perceived degree of certainty as associated with the modals is evident among speakers of Thai, even conceding that the boundary between possibility and probability is often dependent on the speaker’s subjective judgment. For instance, Iwasaki & Ingkaphirom (2005: 137) state that *tôη* signals high confidence; *nâa* and *khuan* signal medium confidence (*nâa* being more ‘subjective’ than *khuan*); and *khon* and *àat* signal low confidence. An anonymous reviewer stated that *nâa*, *àat* and *khon* are widely accepted to indicate a moderate degree of possibility similar to the English *may* and *might*. The judgment may be influenced by the propositional content as well. Minor perceptual disagreements notwithstanding, *cà*-derived futurity markers may be said to signal E-Necessity and E-Possibility by virtue of involving epistemic modals in their combinations.

Another set of *cà*-derived future modals consists of those marking ‘immediate future’ (Smyth 2002: 69; ‘imminent future’ Heine et al. 2017: 10), as exemplified in the following:

- (8) a. *kamlan-cà* immediate future

เรา กำลังจะ กิน ข้าว

raw kamlan-cà kin khâaw

we FUT eat food

‘We are/were about to eat.’

(Smyth 2002: 69, modified)

- b. *cuan-cà* immediate future

เขา จะจมน้ำ แล้ว

khăw cuan-cà com nâam léew

he FUT drown water PFV

‘He is about to drown.’

(Modern Thai, constructed)

¹⁴The lexeme *àat* is also used as an adjective denoting ‘be brave, courageous, bold, be able’; it is likely, though not conclusively, a heteroseme of the modal adverb/auxiliary *àat* ‘probable’.

- c. *kùap-cà* immediate future
 ฉัน ทำ การบ้าน เกือบจะ เสร็จ แล้ว
chǎn tham kaanbâan kùap-cà sèt léew
 I do homework FUT finish PFV
 ‘I will soon finish (almost finished) my homework.’
 (Modern Thai, constructed)
- d. *thêep-cà* immediate future
 ฉัน หิว แทบจะ ตาย
chǎn hīw thêep-cà taay
 I be.hungry FUT die
 ‘I’m so hungry (I feel like) I’m going to die.’
 (Higbie & Thinsan 2002: 90, modified)
- e. *klây-cà* immediate future
 งาน นี้ ใกล้จะ เสร็จ แล้ว
ḡaan ní klây-cà sèt léew
 work this FUT finish PFV
 ‘The job is nearly done.’
 (Higbie & Thinsan 2002: 89, modified)

The examples in (8) involve future markers that also denote that the event being described will occur in the very near future. However, as Smyth (2002: 69) noted with (8a), the use of the future marker is not restricted to referring to a future time in relation to the speech time, but can also be used to refer to a later time in relation to the time of reference in the past. In the literature, the notion of immediacy is often separately labeled as ‘proximative’ or ‘prospective’ (see Heine 1994; Johanson 2017; Nevskaya 2005, 2015; Kuteva et al. 2019, for terminological issues).¹⁵ Example (8a) involves the compound modal *kamlan-cà*, in which *kamlan* is a progressive marker (note that Iwasaki & Ingkaphirom 2005: 149 label *kamlan-cà* as an ‘inceptive’ (i.e., ‘is about to do’) aspect marker). Example (8b) illustrates the use of the immediate future marker *cuan-cà*, which involves the adverb of proximity *cuan* ‘almost, nearly, close to’ (note that Iwasaki & Ingkaphirom (2005: 149) label *cuan-cà* as a ‘pre-inceptive’ (i.e., ‘is almost done’) aspect

¹⁵Johanson (2017) distinguishes prospective and proximative: prospective being an aspect marker signaling a perspective on events foreseen to occur subsequent to some reference time, and proximative being an ‘actionality’ marker referring to a pre-phase (preliminary or preparatory) of a possible event conceived of as imminent or impending, i.e., as ‘about to occur’ (as cited in Nevskaya 2015). For similar issues, see Comrie (1976: 64); Dahl (1985: 111–112); among others.

marker). The closeness in degree or space is used to denote temporal proximity. Similarly, (8c), involving the immediate future *kùap-cà*, makes use of the degree adverb/auxiliary *kùap* ‘almost, nearly, close to’. Likewise, (8d) makes use of the degree adverb *thêep* ‘almost, nearly, barely’, and (8e) makes use of *klây* ‘(be) near, beside’, a polyfunctional exponent in the adjective, adverb, and preposition word class.

Just as *cuan*, *kùap*, *thêep*, and *klây* are synonymous, so are the derived future modals *cuan-cà*, *kùap-cà*, *thêep-cà*, and *klây-cà*. There is no perceived difference in the degree of proximity in these immediate future markers. However, as compared to *kamlan-cà* in (8a), the degree of immediacy signaled by these four, (8b) through (8e), is greater.

There is another, sui generis future marker, i.e., *mák-cà*, the usage of which is exemplified below:

- (9) *mák-cà* future/habitual
 น้ำหวาน มักจะ กิน ขนม ใน รถ
Namwan mák-cà thaan khanǝm nay rót
 [name] FUT eat snack in car
 ‘Namwan would eat (always eats) snacks in the car.’
 (Modern Thai, constructed)

The future marker *mák-cà* is a compound modal involving the degree adverb *mák* ‘frequently, often’. By virtue of the involvement of the full-fledged future marker *cà*, the compound modal *mák-cà* marks the future, and by virtue of the frequency adverb *mák*, it marks the high frequency of repeated occurrence, hence a habitual future. Thus, the event being described is thought to have occurred habitually in the past, at the present time, or in the future. In Modern Thai, the semantic bleaching occurred to such an extent that *mák-cà* is also used for habitual occurrence of past events (as in the English modals *used to* or *would*). This seems to be due to the fact that the speaker’s assessment that some events will occur in the future with regularity is often based on the same events having happened with regularity in the past and in the present. If that is the case, using a future habitual marker for past habitual events is an instance of reasonable extension along the timeline (cf. the English habitual *will* and *would* based on prediction, Quirk et al. 1985: 228–229).

2.3 Adverbials

There are other, less productive, less entrenched, future markers (Supanvanich 1973), i.e., immediate future markers originating from ‘almost, close, nearly’, i.e., *cuan*, *kùap*, *thêep*, and *klây*, as exemplified below:

(10) a. Immediate future

ฝน จวน ตก แล้ว กลับบ้าน กัน

fǒn cuan tòk léew klàp bâan kan

rain FUT fall PFV return home PTCL.HORT

‘It’s going to (almost started to) rain. Let’s go back home.’

(Modern Thai, constructed)

b. เขา เกือบ/แทบ/ใกล้ จม น้ำ แล้ว

khǎw kùap/thêep/klây com náam léew

he FUT drown water PFV

‘He is about to drown.’

(Modern Thai, constructed)

The examples in (10) illustrate *cuan*, *kùap*, *thêep*, and *klây* functioning as future markers. These four lexemes are those that formed compound modals for the immediate future (see 8b through 8e above). In other words, these examples differ from the previously addressed *cuan-cà*, *kùap-cà*, *thêep-cà* and *klây-cà* only in that they do not involve the primary future marker *cà*; the futurity is marked solely by the proximity markers denoting ‘almost, nearly, close’.

One peculiarity of these adverb-based immediate future markers is that they often, but not necessarily, co-occur with *léew*, the perfective aspect particle, also functioning as a sentence ending particle of diverse discourse functions (note that 10a is bi-clausal (or bi-sentential), the first one ending with *léew*), which originated from the verb *léew* ‘finish, be done’ (Iwasaki & Ingkaphirom 2005: 149).¹⁶ The combination of the perfective *léew* with the proximity degree markers signals that something will not only occur soon but will also be completed soon, and thus creates the sense of definite and complete occurrence of the event within a short period of time.

Treating temporal adverbials as future markers, as Supanvanich (1973) does, raises an important issue that warrants mention, as an anonymous reviewer pointed out. According to the reviewer, adverbials modify actions with respect to their degree of progress, but the described actions may not be actualized. As researchers have observed, futurity is never a purely temporal concept but it necessarily includes an element of prediction or some related modal notion (Lyons 1977: 677; see also Salkie 2010: 1787). There is always an inherent uncertainty about whether an event will take place or not (Dahl 1985, 2000; Jäggi et al. 2022). Thus, a future marker does not guarantee the occurrence of a described event or

¹⁶The immediacy effect created by *léew* is evident in the fact that when the primary future marker *cà* occurs with *léew*, the sentence means that the event being described will occur very shortly.

the presence of a described state of affairs in the future; it only signals that an event or state is, unless stated otherwise, to come about at a time later than the speech time or referenced time. The detachment of a future-marked event from its actualization in reality is inherent in the notion of future, and this trait closely links the future marker to irrealis (Noss 1964; Soithurum 2010: 110). Noss (1964: 165), in his exposition of *cà*-, states that the meaning of *cà*- is ‘hypothetical situation or putative action.’ In this regard, the analysis by Iwasaki & Ingkaphirom (2005: 123) of *cà* as a marker of ‘challengeability’ (Givón 1982) is particularly noteworthy. In other words, a future event is assumed to be challengeable by virtue of not having been realized and thus may be marked by *cà*. Such an analysis can shed light on the usage of *cà* for non-temporal marking, the discussion of which is beyond the immediate scope of the present paper (see Iwasaki & Ingkaphirom 2005: 126–127 for more exposition).

The issue of whether or not the forms marking the degree of progress may be regarded as future markers is debatable. Two points are relevant to this issue: (i) the Mandarin Chinese temporal lexeme *yào* ‘immediately’ (as well as ‘want’) functions as a ‘lexical futurity’ marker (Li & Thompson 1989: 175), and (ii) the general notion of futurity may include temporal adverbials, which are lexical, as well as proximative and prospective adverbials, which are often on the boundary of lexis and grammar. In this light, the adverbials that denote temporal proximity addressed here may well be regarded, at least, as ‘peripheral’ markers of futurity.

2.4 Particles

Other linguistic forms have additionally been claimed to be future markers in Thai. For instance, Supanvanich (1973) asserts that certain sentence ending particles are future markers, e.g., *lá*, *ná*, and *lá-ná*. These future markers are subject to controversy and, even if qualified as such, are undoubtedly marginal forms. Such uses are exemplified in the following:

(11) Immediate future

ฉัน นอน ละ / ูะ / ละูะ

chǎn nɔɔn lá / ná / lá-ná

I sleep PTCL

‘I (will) sleep now.’

(Supanvanich 1973)

The example in (11) involves the sentence ending particles *lá*, *ná*, and *lá-ná*. The primary function of these particles, when used in any sentence, is to draw the addressee’s attention to the information being provided and indicate this information as noteworthy. The notion of futurity associated with these forms seems

to be only pragmatically inferred, i.e., not yet grammaticalized. In other words, when the example sentence (11) is spoken, and the speaker is not yet in bed, the addressee, noticing the discrepancy, reconciles it by inferring that the proposition (*Chǎn nǎon* ‘I sleep’), presented as a piece of ‘noteworthy’ information (as indicated by *lá/ná/lá-ná*), is describing a situation ‘as good as the present’, hence the immediate future. Put another way, in (11) above, there exists a gap between the speaker’s alleged claim (i.e., ‘I am sleeping’) and the hearer’s perceived reality (i.e., ‘You are not sleeping now’), which triggers the pragmatic inference (‘You will go to sleep immediately’). The motivation behind describing a future event as one that is presently occurring is conceptually straightforward, and is classifiable as an instance of hyperbole (cf. ‘That does it! I’m leaving’ Quirk et al. 1985: 215).¹⁷ This pragmatic inference seems to have triggered the functional reanalysis of the sentence ending particles as immediate future markers, or at least as forms with a future connotation. It is to be noted that this inference of future time reference is triggered by only these sentence ending particles out of numerous particles in Modern Thai. Therefore, this is different from the use of the simple present tense for future, which is widely attested across languages (cf. Quirk et al. 1985: 182–183 for English). Incidentally, the variations by means of the three sentence ending particles do not create any significant differences in meaning, but most Thai speakers understand that when the longer particle (*lá-ná*) is used instead of the shorter ones (*lá* and *ná*), the proposition carries stronger illocutionary force.

3 Discussion

In the preceding section, we have seen how Thai future markers are used. Based on the description, we will discuss a few issues involved in their development that bear theoretical significance, e.g., morphosyntactic motivation, conceptual motivation, and grammaticalization parameters.

3.1 Morphosyntactic Motivation

The morphosyntactic motivation behind the development of Thai futures comprises two primary factors, one typological, and the other idiosyncratic. The typological factor relates to pragmatic orientation and verb serialization, and the idiosyncratic factor relates to the preference of creating polylexemic units in lexicalization and grammaticalization.

¹⁷The role of hyperbole in grammaticalization has been reported in a series of works by Rhee (2003, 2009, 2022) on the Korean language.

Most functionalists hold that languages differ in terms of the degree to which they utilize pragmatic factors in their linguistic representation. For instance, Huang (2000: 276) asserts that ‘Some languages are more pragmatic than others’ (for similar positions, see Li & Thompson 1976, 1979; and on Thai, see Bamgroongraks 1987; see also O’Grady 2005, 2008, for the ‘emergentist’ approach to linguistic structure). Thai lacks many grammatical forms that are found in many languages, notably linking words. Thus, the semantic and syntactic relationships between constituents are often identified from the linguistic and situational context. This pragmatic orientation is related to the proliferation of verb serialization, in which verbs are strung together *asyndetically*. Diller (2006: 160), while illustrating an overload of verb-based detail in informal Thai, presents an example of a seemingly extreme case: ‘Go. Hurry exit go buy return come finish peel pass’ (for ‘Then hurry off and buy some. When you get back, peel them – get on with it’). The serial verb construction is so susceptible to grammaticalization that it is often regarded as the ‘seed’ of grammaticalization (cf. DeLancey 1991: 15–16). Indeed, all serial verb languages seem to develop exponents of grammatical functions out of serial verb constructions by way of reanalysis (Bynion 1985), or of pragmatics, i.e., interaction between conceptualization of world events and iconicity in the language (cf. Bruce 1988; DeLancey 1991; Lord 1995; Rhee 1996; Aikhenwald & Dixon 2006 and works therein). With the prevalence of verb serialization, Thai exhibits numerous instances of grammaticalization from serialized verbs (Vis 1978; Thepkanjana 1986; Diller 2006; Park 2011).

The development of the primary future tense marker *cà* (and its older *càk*), which originated from a verb, seems to have occurred in verb serialization. The likely lexical origin of the marker is the Sino-Thai lexeme *càk* ‘to know, recognize’, which survives in contemporary Thai as *rúucàk* ‘to know a person, place, etc.’ (see Section 2.1 above and Section 3.2 below). Thus, the source construction of the future marker is, for example, [I + know + return + home], from which the reinterpreted construction [I + will + return + home] emerged (Note that Thai is a head-initial language and the auxiliary precedes the verb).

Secondly, Thai has a strong idiosyncratic preference for the creation of polylexemic or polysyllabic units in lexicalization and grammaticalization (Note that native Thai lexemes are predominantly monosyllabic, Smyth 2002: 1). In other words, Thai often resorts to recycling a single lexeme to create similar or slightly modulated meanings by reduplicating the base form, reinforcing form and meaning by means of the juxtaposing of synonyms, and/or recruiting additional forms, such as particles, adverbs, etc. (cf. ‘polysemy strategy’ Khammee & Rhee 2022; Rhee & Khammee 2022; Thirakunkovit & Rhee 2023). This preference of ‘addition’ is also applicable to the sentence level; various sentence ending particles are

(often multiply) added to the end of the sentence, with the primary function being to soften the illocution and make the assertion or statement more polite. The speech style of ending a sentence without a sentence-final particle is stigmatized as *mây-mii-hǎaŋ-sǎaŋ* ‘having no tail of sound’ (Khammee et al. 2024).

The preference for creating polylexemic or polysyllabic units by means of the above-mentioned strategies is responsible for the creation of several *cà*-derived future markers, e.g., *cà-tôŋ*, *khuan-cà*, *nâa-cà*, *àat-cà*, *khon-cà*, *kamlan-cà*, *cuan-cà*, *khàp-cà*, *thêep-cà*, *klây-cà*, *mák-cà*, etc. (see Section 2.2 above). When multiple modals occur in a single sentence, they tend to occur next to each other in the pre-verbal slot, and thus their positional closeness, as well as the general tendency to create polylexemic, polysyllabic units, is likely to increase the bond between them (hence, *cà-tôŋ* ‘will’ (< ‘will-must’)). Similarly, adverbials often occur next to the future marker. In general, there is a small group of adverbials that occur sentence-finally (but before sentence ending particles), e.g., *samǎə* ‘always’, *bôy* ‘often’, and *rúay* ‘all the time’ (Iwasaki & Ingkaphirom 2005: 103), but others are more positionally flexible and often occur at the pre-auxiliary position. Hence, the habitual future *mák-cà* ‘will’ (< ‘frequently-will’), the immediate future *cuan-cà* ‘be about to, will’ (< ‘almost-will’), etc.

Apart from the two primary factors elaborated above, a mention is in order with respect to the immediate future markers that are heterosemous with degree adverbs, i.e., *cuan* ‘frequently’, *khàp*, *thêep*, and *klây* (see Section 2.3 above). These future markers are less established, as evident in the fact that few researchers include them in the inventory of future forms (e.g., Supanvanich 1973). The syntagmatic position of degree adverbials is flexible, but the pre-verbal position is favored, because they are modifiers of the verb or verb phrase. Thus, in this configuration, degree adverbials occupy the same position as the auxiliaries, including future markers such as *cà*, which, coupled with conceptual motivation (see below), seems to have triggered the functional reinterpretation and structural reanalysis of the degree adverbials into future auxiliaries.¹⁸

3.2 Conceptual Motivation

Semantic change in grammaticalization is not random but, rather, is motivated (‘the source determination hypothesis’ and ‘the universal path hypothesis’; Bybee et al. 1994). Conceptual motivation and the gradient nature of linguistic change lead to the formation of a conceptual network of lexemes and grammatical forms (Heine et al. 1991; Nuyts 1992; ‘semantic map’ van der Auwera & Plungian 1998; Croft 2001; Narrog 2009).

¹⁸See Section 2.2, however, for unsolved morphosyntactic positionality issues.

We have noted that the source lexeme of the Thai primary future marker *cà* is likely the Sino-Thai verb *càk* ‘to know, recognize’ (see Section 2.1 above) and that around the time when it became grammaticalized into a future auxiliary in the Sukhothai period (1238–1347 CE), its meaning became ‘to intend’ (see Section 2.2 above). Crosslinguistic research shows that the future develops from diverse sources, e.g., Change-of-state, ‘come to’, ‘go to’, ‘love’, B-Necessity, D-Necessity, H-Possessive, ‘take’, ‘then’, ‘tomorrow’, Venitive, and ‘want’ (Kuteva et al. 2019). Since B-Necessity denotes ‘intention’, the lexical source of the Thai path conforms to the crosslinguistic pattern (also the path proposed by Bybee et al. 1994). Furthermore, the lexical source ‘know’ is known to develop into PI-Possibility (Ability) and further to Habitual or Permission, or to Evidential or Experiential (Kuteva et al. 2019). Mandarin Chinese has no grammatical tense, but the verb *hui* ‘know’ can indicate futurity (Bybee et al. 1994: 265; Li & Thompson 1989: 183). Similar to this grammaticalization scenario, the Medieval Chinese verb *jie* ‘know, know of’ developed into the auxiliary of PI-Possibility (Ability), which describes a notion closely related to the future (Long et al. 2012). The Thai future marker *cà* marks Possibility (E-Possibility), closely related to PI-Possibility (Ability), as well as Necessity (i.e., E-Necessity, for Probability; D-Necessity, for Obligation; and B-Necessity, for Intention), often with the help of co-occurring forms. The states of affairs in Thai fit with the observed crosslinguistic patterns.

It is noteworthy that once a lexical form or a construction begins to mark future tense, the marker, in most cases, seems to further acquire diverse modal functions, such as Necessity, Certainty, Probability, etc., even though the focus may be selective and different. For instance, the Thai future acquired the modal function of Indeterminacy, which is likely to have derived from the characteristic of the future as inherently indeterminate and uncertain.

The adverbials of proximity denoting ‘almost, nearly’, either alone or with the primary future *cà*, have developed into future markers (see Section 2.2 and Section 2.3 above). The development must have been enabled by the general proximity (typically spatial and degree) metaphorically projected to temporal proximity. The original semantics still persists in the grammaticalized markers in that these are immediate future markers (rather than remote future or general future) and that their functions are largely restricted to marking immediate future, not extended to other modal meanings.

We also noted that some sentence ending particles have developed into future markers, albeit marginally (see Section 2.4 above). Considering that the primary function of these markers is to signal that the speaker is ‘informing’ the addressee by providing noteworthy information, we argued that the addressee’s attempt to

fill the gap between the information and reality is responsible for the emergence of the function of marking immediate future.

All these conceptual considerations and their interrelatedness may be sketched, as in Figure 1.

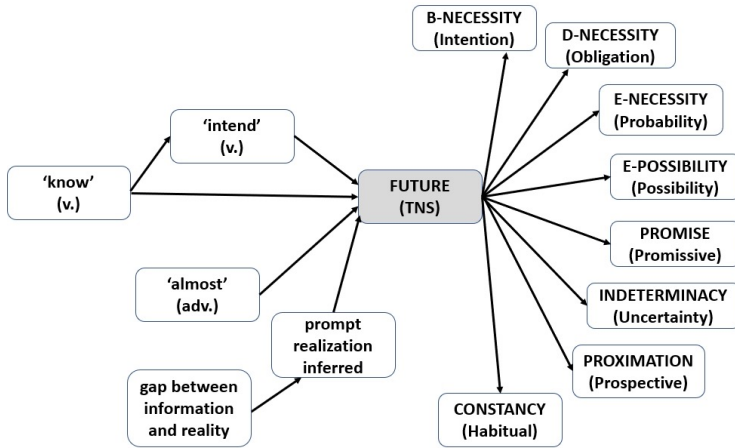


Figure 1: Conceptual network of Thai future markers

3.3 Grammaticalization Parameters

A brief mention to Thai documentation is in order before we discuss diachronic development. The documentation of the Thai language has a rather shallow historical depth. Haarmann (2012), Leppert (1992), and others state that it is possible that the early literature of the Thai people may have been written in Chinese, but no historical record of the Siamese thus far refers to this earlier literature. The Thai alphabet was established in the 13th century CE, and the earliest literature dates back to the Sukhothai period (13th to mid-14th century), chiefly surviving in the form of stone inscriptions (e.g., the King Ramkhamhaeng inscription of 1292). Classical literature dating from the Ayutthaya period (1351–1767) is written in verse but most of it was lost in the 18th century in the military conflict with Burma (Myanmar). Thus, the extant literature is recent, mostly dating back to the 19th century. One obvious implication of this shallow historical depth in grammaticalization studies is that the developmental trajectories of grammatical forms cannot be easily or conclusively established. Thus, following the tradition of researching languages with little or no historical documentation (Traugott & Heine 1991: 6), we can project diachronic grammaticalization to synchronic data

and, conversely, synchronic variations to the reconstruction diachronic developmental paths.

The relative paucity of historical data notwithstanding, we will review the grammaticalization scenarios from the available resources with respect to the four grammaticalization parameters suggested in Kuteva et al. (2019), i.e., desemanticization, extension, decategorialization, and erosion.

3.3.1 Desemanticization

Also commonly called ‘semantic bleaching’ in figurative terms, desemanticization refers to the loss in meaning content. As briefly alluded to in the preceding exposition, the source lexeme *càk* has undergone desemanticization to such an extent that most speakers of contemporary Thai are unable to recognize the connection between the verb *càk* and the future marker *cà*. The implication of cognizance, i.e., ‘know’, is nearly completely lost and its trace is only visible in the verb *rúucàk* ‘to know a person, place, etc.’ Even as a future marker, *cà* does not make a substantial semantic contribution to the *cà*-compounds. For instance, the semantic difference between the compound and non-compound is often imperceptible (e.g., *àat* ‘might’ and *àat-cà* ‘might’; *khon* ‘possibly’ and *khon-cà* ‘would’; *nâa* ‘probably’ and *nâa-cà* ‘likely’; *khuan* ‘should’ and *khuan-cà* ‘should’; *kàap* ‘almost’ and *kàap-cà* ‘almost’; etc.).

On the other hand, the meanings of general degree and spatial proximity as associated with the proximity adverbs denoting ‘almost’ have been bleached, but the bleaching is not substantial because the immediate future is still based on the notion of temporal proximity. The sentence ending particles, by virtue of having no ‘semantic’ content, cannot be discussed in this regard.

3.3.2 Extension

Also known as context generalization, extension refers to the use of a form in new contexts. With respect to the future *cà*, the source lexeme was a verb denoting the presence of knowledge, which later denoted intention, and after grammaticalization the form has become an auxiliary of the future. Through grammaticalization, the usage context has been extended because there is no more semantic constraint and the form can modify a proposition (i.e., a clause).

Similarly, the degree adverbs that, before grammaticalization, could be used as modifiers of verbal, adjectival, and predicate constituents, can now modify propositions, a clear instance of usage context extension.

The sentence ending particles that could be used as modifiers of a clause and function as a discourse operator (as markers of the speaker’s diverse stances) can

now be used as modifiers of a clause with respect to the temporal dimension of the proposition, which also constitutes an instance of usage context extension.

3.3.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Decategorialization typically involves, but is not restricted to, the primary categories. We will look at the source verb *càk* and the adverbs *cuan*, *kàap*, *thêep*, and *klây*.

The verb *càk*, by virtue of being a verb, could carry diverse verbal features, such as degree, quality, frequency, and manner modification, e.g., ‘fast’, ‘well’, ‘everyday’, ‘diligently’, ‘thoroughly’, etc. However, after its grammaticalization into a future marker, the form cannot be marked by such modifiers. The ability to c-select a theme as its complement, a feature of a transitive verb, has also been lost.

Similarly, the proximity adverbials, before their grammaticalization into future markers, could function as modifiers of verbs, adjectives, and predicates. After grammaticalization, such adverbial features are lost and can function only as modifiers of a clause. Thus, the grammaticalization of future markers in Thai clearly shows decategorialization.

3.3.4 Erosion

Also known as phonetic reduction, erosion refers to the loss in phonetic substance. Even though most grammaticalizing forms across languages undergo phonological reduction, often by virtue of increased frequency, Thai future markers are not eroded to a significant level, a characteristic of an isolating language (cf. Matisoff 1991; Bisang 1996, 2011; Enfield 2009; Ansaldi et al. 2018). For instance, the verb *càk* did not lose any substance in its development into the future marker *càk*. As it developed into the future marker *cà*, some minimal loss has occurred.

Similarly, the proximity adverbials have not undergone phonological erosion either, in the course of their development into future markers. Their syntagmatic properties as well as their phonological shapes (including tonal features) are retained even after the grammaticalization. The only change that occurred is their functional reinterpretation. The same is applicable to sentence ending particles: they retain all syntagmatic and phonological properties after their grammaticalization. The grammaticalization only lies in the reinterpretation of a discourse-pragmatic marker into a tense-marking modal auxiliary.

There is one noteworthy aspect in the grammaticalization of future markers in Thai, which may be extendable to Thai grammaticalization in general. It relates to the reverse of erosion, i.e., reinforcement. As indicated in Section 3.1, Thai has a strong tendency to reinforce a linguistic (particularly grammatical) form. The common strategies recruited in the realization of this inclination include reduplicating the base form, reinforcing form and meaning by means of juxtaposing synonyms, and recruiting additional forms, such as particles, adverbs, etc. (see Section 3.1 above). For instance, in their research on Thai allatives numbering 26, Rhee & Khammee (2022) report that there are many forms that recycle the base form by reinforcing it with semantically similar forms, e.g., *hây* ‘to/for’, *hây-kêe* ‘to/for’, *hây-kàp* ‘to/for’ all involving *hây* ‘to/for’; *con* ‘to’, *con-thŭŋ* ‘to’, *cùap-con* ‘to’, *con-kràthân-thŭŋ* ‘to’, *tràap-con* ‘to’, *tràap-con-thŭŋ* ‘to’, all involving *con* ‘to’, etc. (see also Khammee & Rhee 2025 for a similar analysis). Similarly, Thirakunkovit & Rhee (2023), in their research on Thai intensifiers, note that out of 23 intensifiers as many as 14 of them involve *mâak* ‘very, very much’, the primary intensifier in contemporary Thai, e.g., *mâak* ‘very’, *mâak-maay* ‘very’, *mâak-maay-kàay-kɔŋ* ‘very’, *mâak-kəən-pay* ‘very, too much’, *mâak-con-kəən-pay* ‘very, too much’, etc. These forms, though composite in makeup, are perceived as ‘univerbated’ units (Lehmann 2015) by Thai speakers. Even though further research is required to reach a solid generalization, Thai seems to favor semantic and phonological reinforcement to increase the salience, especially as a grammaticalizing form becomes semantically weak or loses novelty through frequent use.

4 Summary and Conclusion

Contemporary Thai has a number of forms that mark futurity. In addition to the primary marker *cà* and its older form *càk*, there are secondary markers of futurity, e.g., proximity adverbials denoting ‘almost’ and sentence ending particles signaling informative noteworthiness, which are used either alone or in combination with other forms. An investigation of their morphosyntactic development reveals some influence of the typological and idiosyncratic features of Thai, such as a high level of pragmatic orientation, extensive verb serialization, and preference for polylexemic units in lexicalization and grammaticalization, among others. Their semantic developments are also well motivated in terms of conceptual relatedness between the source forms and the grammaticalized forms. A review of grammaticalization scenarios with reference to grammaticalization parameters, e.g., desemanticization, extension, decategorialization, and

erosion, shows that the Thai future markers have been grammaticalized to varying degrees. One peculiarity of Thai future grammaticalization is that Thai favors semantic and phonological reinforcement to increase the salience, especially as a grammaticalizing form becomes semantically weak or loses novelty through frequent use, thus countering semantic bleaching and formal erosion.

Abbreviations

FUT	future	PFV	perfective
HON	honorific	PRS	present
HORT	hortative	PROG	progressive
M	masculine	PTCL	particle
NMLZ	nominalizer	SG	singular

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